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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Course : Broad Discipline of COMP-61431

Subject : COMP4422 Computer Graphics

Group : 1011

Session : 2023 / 2024 Semester I

Date : 18 Dec 2023 Time : 19:00-21:00

Time Allowed: 2 Hours Subject Lecturer: Prof. YOU Jane

This question paper has 3 pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates:

1. This is a close book examination which consists of 8 questions.
2. Students should attempt ALL questions.
3. Marks for each question are shown next to the question. Total marks = 100.

Do not turn this page until you are told to do so!

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Question 2 ***[Total = 10 marks]***

Please fill in the blanks of the following statements (*1 mark for each blank*).

In computer graphics, the process of removing lines or portions of lines outside of an area of interest is called _____. Such an area of interest is referred to as _____ in 2D and _____ in 3D. Cohen-Sutherland line clipping algorithm divides a 2D spece into _____ parts, of which each part is assigned a _____bits code. Such a code is also called _____. The region code of the middle part (viewport) is _____. The algorithm _____, _____ or _____ the line based on where the two endpoints are.

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Question 3 ***[Total = 10 marks]***

Match the following terms with the appropriate descriptions (*1 mark for each item*)

Item		Description	
1	Viewport	A	A method of mapping three-dimensional points to a two-dimensional plane
2	NDC	B	The process of creating realistic graphics scenes from geometric data
3	Rendering	C	Normalized Device Coordinate System
4	Wireframe model	D	Intensity of a chromatic light
5	Orthographic projection	E	The process of creating realistic graphics scenes from geometric data
6	Brightness	F	Degree to which a pure color is diluted by white
7	3D projection	G	Color
8	Saturation	H	A rectangular area specifies on a display device to which a window is mapped
9	Rendering	I	Objects represented by a set of points (vertices) and a set of edges.
10	Hue	J	A type of parallel projection where the projection rays are perpendicular to the projection plane.

Question 4 ***[Total = 20 marks]***

Please describe the following terms or operations briefly (*5 marks for each questions*).

1. 4-neighborhood and 8-neighborhood

2. the basic data structure of wireframe model used for 3D object representation

3. parallel projection and perspective projection

4. the difference between Gourand Shading method and Phong Shading method

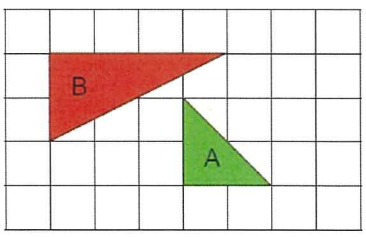
Question 5 ***[Total = 20 marks]***

Given the three matrices **A**: translation along **x** axis by 4, **B**: rotation 90 degrees around the origin and **C**: a non-uniform scaling with 2 in x and 3 in y. Please answer the following questions:

a) Give the (3x3) matrix form of each of **A**, **B** and **C**. *(5 marks)*

b) Given the point P = (1, 2, 1) in homogeneous coordinates, what is the location of P0 = **CAB**P? *(5 marks)*

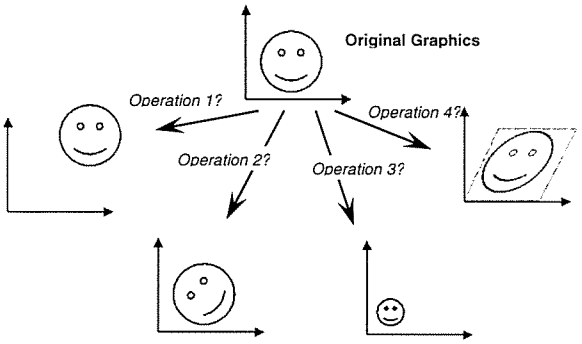
c) Describe the transform needed to transform the triangle from A to B in the Figure below, by a set of matrices. *(10 marks)*



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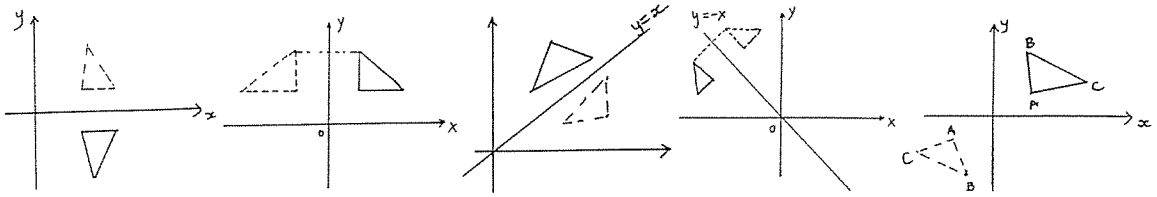
Question 6 ***[Total = 10 marks]***

(a) Identify the required transform operations marked in the following diagram to generate new graphics from the original sample. *(5 marks)*



(b) Derive the transformation matrices of the following specified reflections *(5 marks)*

(1) about x-axis *(2) about y-axis* *(3) about y-x* *(4) about y=-x* *(5) about the origin*

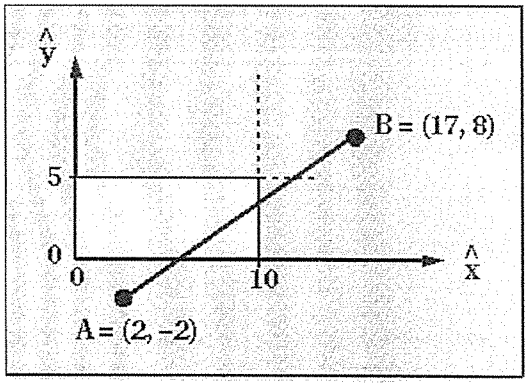


Question 7 ***[Total = 10 marks]***

What are the parameters required to specify the view for a perspective projection? When the eye is located at (0,0,-6) and the projection plane is the xy plane (z=0), show on a diagram the projector from the point (2,6,8) and the projection of this point. Explicitly derive the projection of the given point (2,6,8) using the simple perspective projection matrix.

Question 8 ***[Total = 10 marks]***

Use the Cohen-Sutherland clipping algorithm to clip line AB to the 10x5 window shown in the figure. Write down all your steps.



==========THE END==========